Flashing the Official SUPER Clean System

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Installing the Official Clean System from a USB Drive

The NVIDIA JetPack download page states that Jetson Orin NX no longer provides an SD card image; instead, the unified Jetson installer ISO is used to create a USB drive. The JetPack 7.2.1 / Jetson Linux R39.2.1 ISO uses the Super configuration by default for Orin NX. The procedure below applies to the **official NVIDIA Orin NX with NVMe as the target**. For third-party SUB carrier boards, or for modules that are actually Orin NX, you must first obtain the vendor's confirmation that R39.2.1 QSPI/carrier-board support is available — do not simply reuse the official board configuration.

This is a reinstall, not an in-place upgrade that preserves data. The installer erases the selected target drive; back up the system, user files and configuration on the NVMe first. Creating the installation drive also erases the USB drive. The official clean system does not retain the preinstalled examples from the original course. Use a stable power supply throughout, and never interrupt power while the QSPI is being written.

1 Confirm the Hardware and Boot Prerequisites

This procedure has two different write targets: the on-board QSPI holds the UEFI/boot firmware, while the NVMe holds Jetson Linux. **Complete the required QSPI update first, then install to the NVMe.** If the official Orin NX already has a JetPack 6.x-generation UEFI (for example, the boot screen shows 36.x), you can create and boot the ISO USB drive directly. When the QSPI update prompt appears, you must press `Y` within 30 seconds and wait for both capsule updates to complete. If you do not press `Y` and the prompt times out, the installer may keep running but can still fail late in the NVMe installation. If the device is already on a compatible version and no update prompt appears, simply continue with the installer flow.

If the UEFI version is older than JetPack 6.x, the system cannot boot from the ISO USB drive, or the update fails after pressing `Y`, stop the NVMe installation and consult the official JetPack 6.x firmware upgrade path and QSPI-only flashing. The latter requires an x86-64 Linux host, Recovery mode and a USB data connection; the actual board configuration must match the module and carrier board, and third-party SUB carrier boards cannot simply reuse the official commands. Do not use an unsigned public ISO on a device with Secure Boot enabled.

2 Create the Installation USB Drive

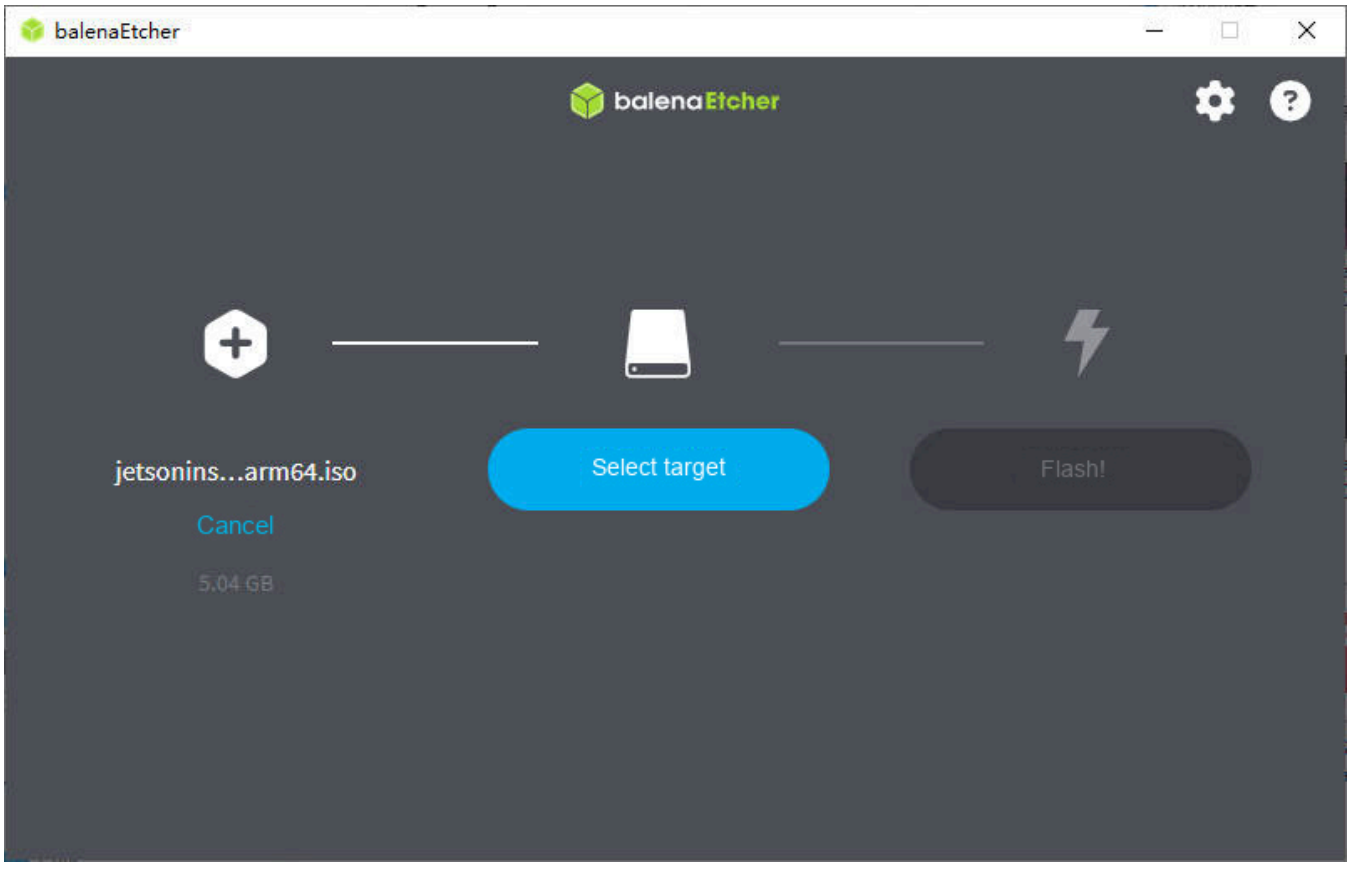
1. Prepare a Windows, macOS or Linux computer (at least 25 GB of free space) and a **16 GB or larger** USB drive. Check the USB drive letter carefully — its data will be overwritten.
2. Download the **JetPack 7.2.1 ISO** from the NVIDIA JetPack download page; the current file is `jetsoninstaller-r39.2.1-2026-08-07-18-30-47-arm64.iso`, about 5 GB. If the version or file name on the download page changes, follow what the page actually publishes.

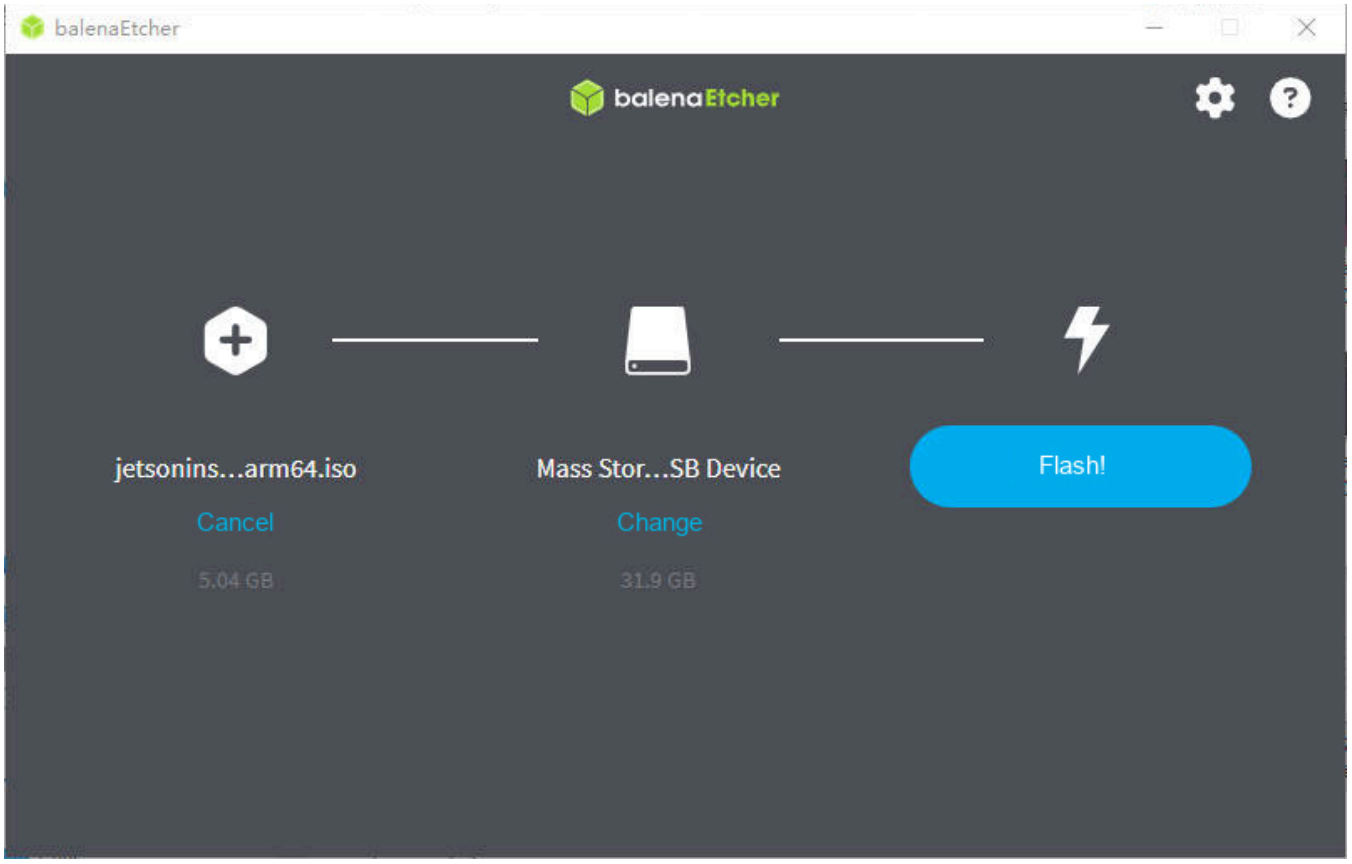


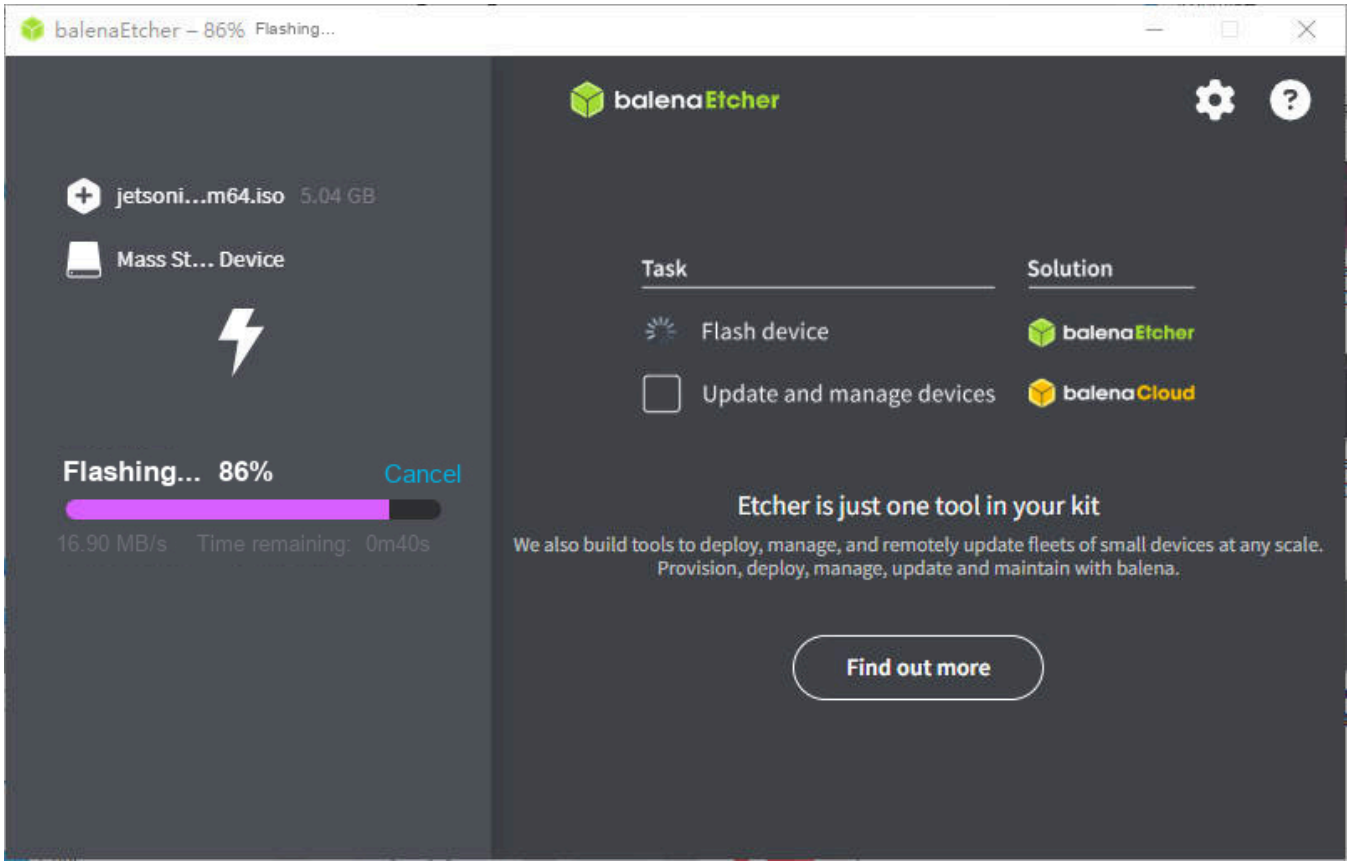
3. Use balenaEtcher to select the ISO, select the target USB drive and run Flash, then safely eject the drive once the writing and verification are complete.

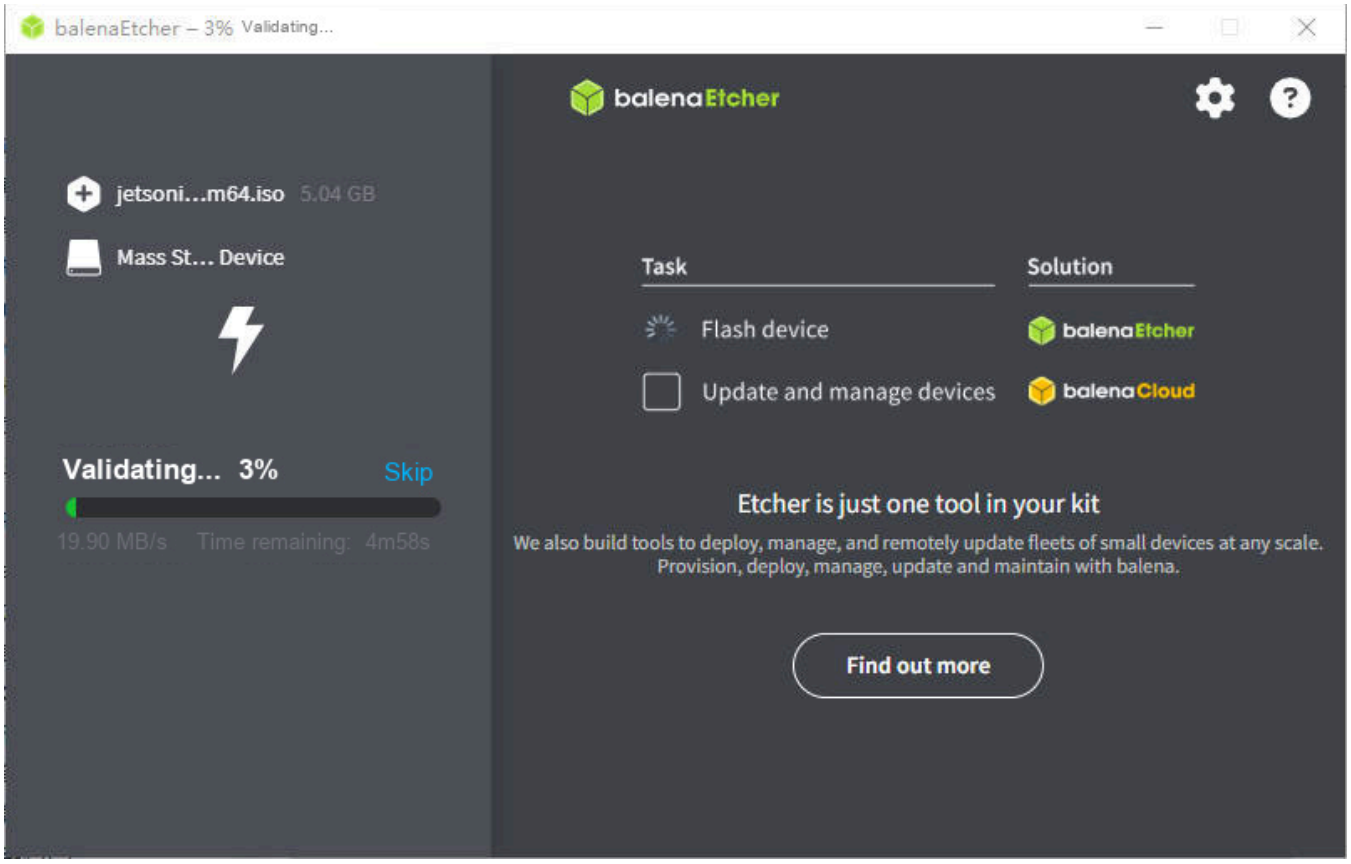
> Note: Copying the `.iso` file with a file manager is not a substitute for flashing it to a drive, and a microSD card inserted into a reader cannot replace a USB drive. NVIDIA explicitly requires writing the ISO to a **USB flash drive**, not to a microSD card.

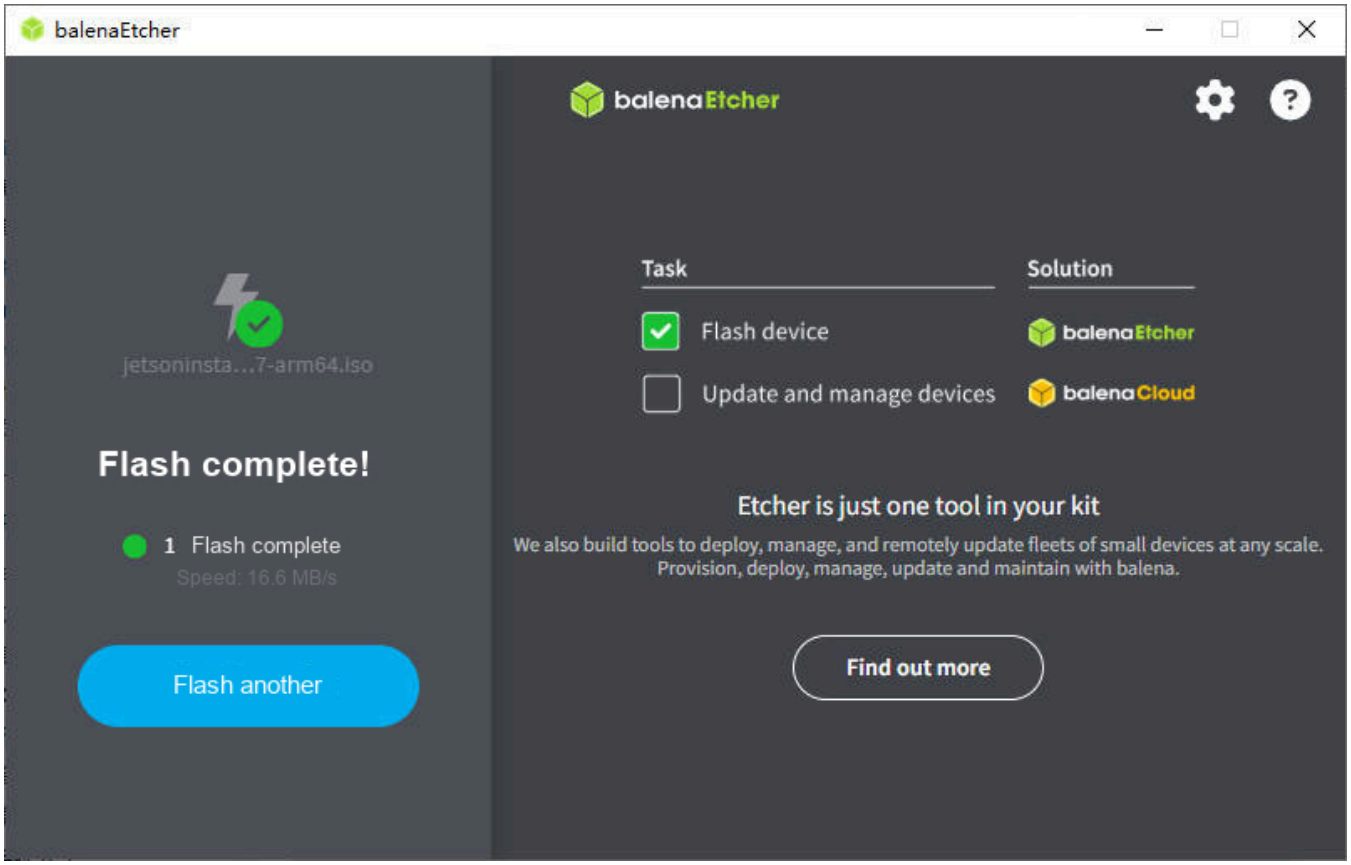

balenaEtcher
Flash. Flawless.
Flash OS images to SD cards & USB drives, safely and easily.
Select drive
Download Etcher
We use cookies
We use cookies to enhance your browsing experience and analyze our traffic. By clicking "Accept All" you consent to our use of cookies. Read more about our privacy policy.
Customize
Reject all
Accept all
RESOURCES


balenaEtcher
balenaEtcher
jetsonins...arm64.iso
Cancel
5.04 GB
Select target
Flash!


balenaEtcher
balenaEtcher
jetsonins...arm64.iso
Cancel
5.04 GB
Mass Stor...SB Device
Change
31.9 GB
Flash!

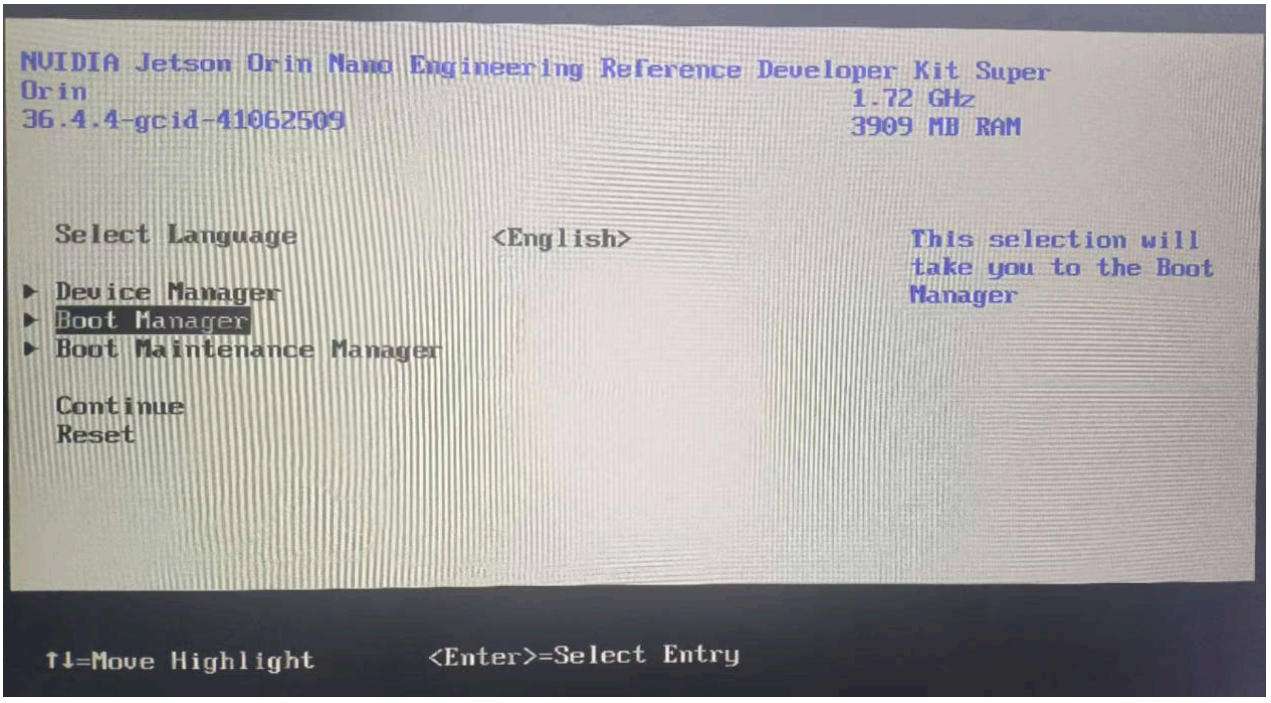

balenaEtcher – 86% Flashing...
balenaEtcher
jetsoni...m64.iso 5.04 GB
Mass St... Device
Flashing... 86%
Cancel
16.90 MB/s Time remaining: 0m40s
Task
Solution
Flash device
balenaEtcher
Update and manage devices
balenaCloud
Etcher is just one tool in your kit
We also build tools to deploy, manage, and remotely update fleets of small devices at any scale.
Provision, deploy, manage, update and maintain with balena.
Find out more

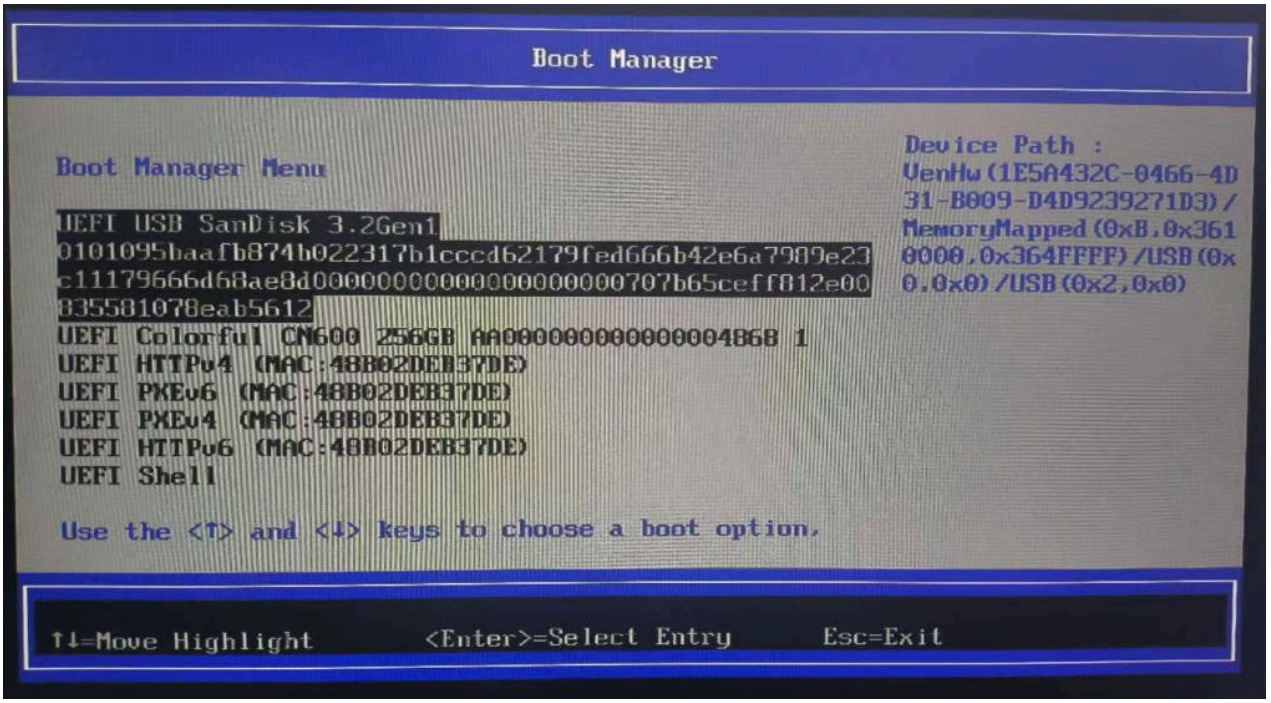

balenaEtcher – 3% Validating...
balenaEtcher
jetsoni...m64.iso 5.04 GB
Mass St... Device
Validating... 3%
Skip
19.90 MB/s Time remaining: 4m58s
Task
Solution
Flash device
balenaEtcher
Update and manage devices
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Etcher is just one tool in your kit
We also build tools to deploy, manage, and remotely update fleets of small devices at any scale.
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Find out more


balenaEtcher
balenaEtcher
jetsoninsta...7-arm64.iso
Flash complete!
1 Flash complete
Speed: 16.6 MB/s
Flash another
Task
Solution
Flash device
balenaEtcher
Update and manage devices
balenaCloud
Etcher is just one tool in your kit
We also build tools to deploy, manage, and remotely update fleets of small devices at any scale.
Provision, deploy, manage, update and maintain with balena.
Find out more

3 Install from the USB Drive to the NVMe

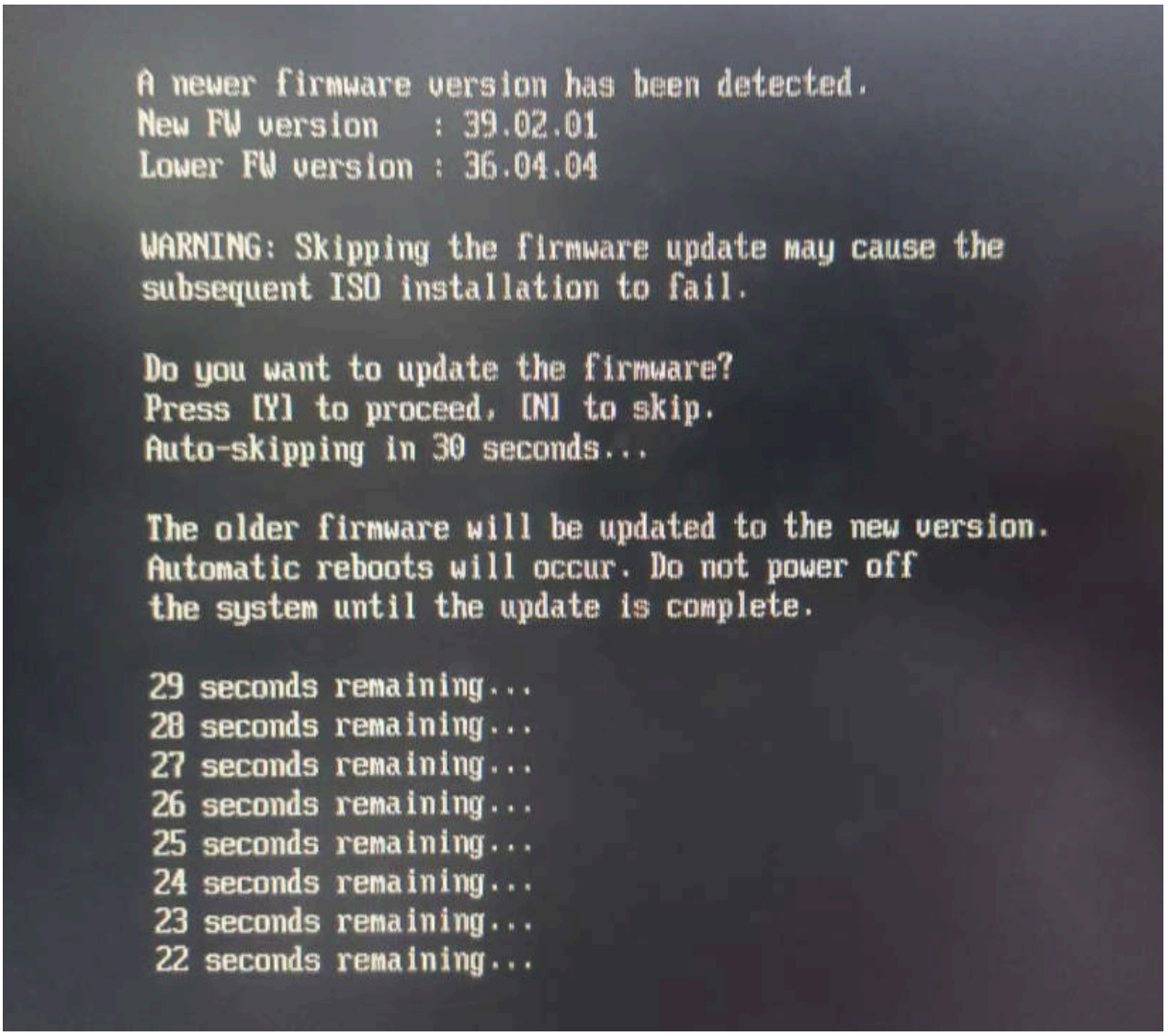
1. With the power off, confirm that the NVMe drive is installed; connect the display and a USB keyboard, insert the installation USB drive, and only then apply power. The whole process takes more than twenty minutes, so make sure the board stays powered on the entire time and keep metal objects away from it to avoid accidents.
2. Normally the device boots from the USB drive. If the installer does not start, press `Esc` repeatedly when the NVIDIA logo appears to enter the UEFI `Boot Manager`, and select the USB drive to boot from; adjust the boot order if needed, then save and exit. See the screenshots below:

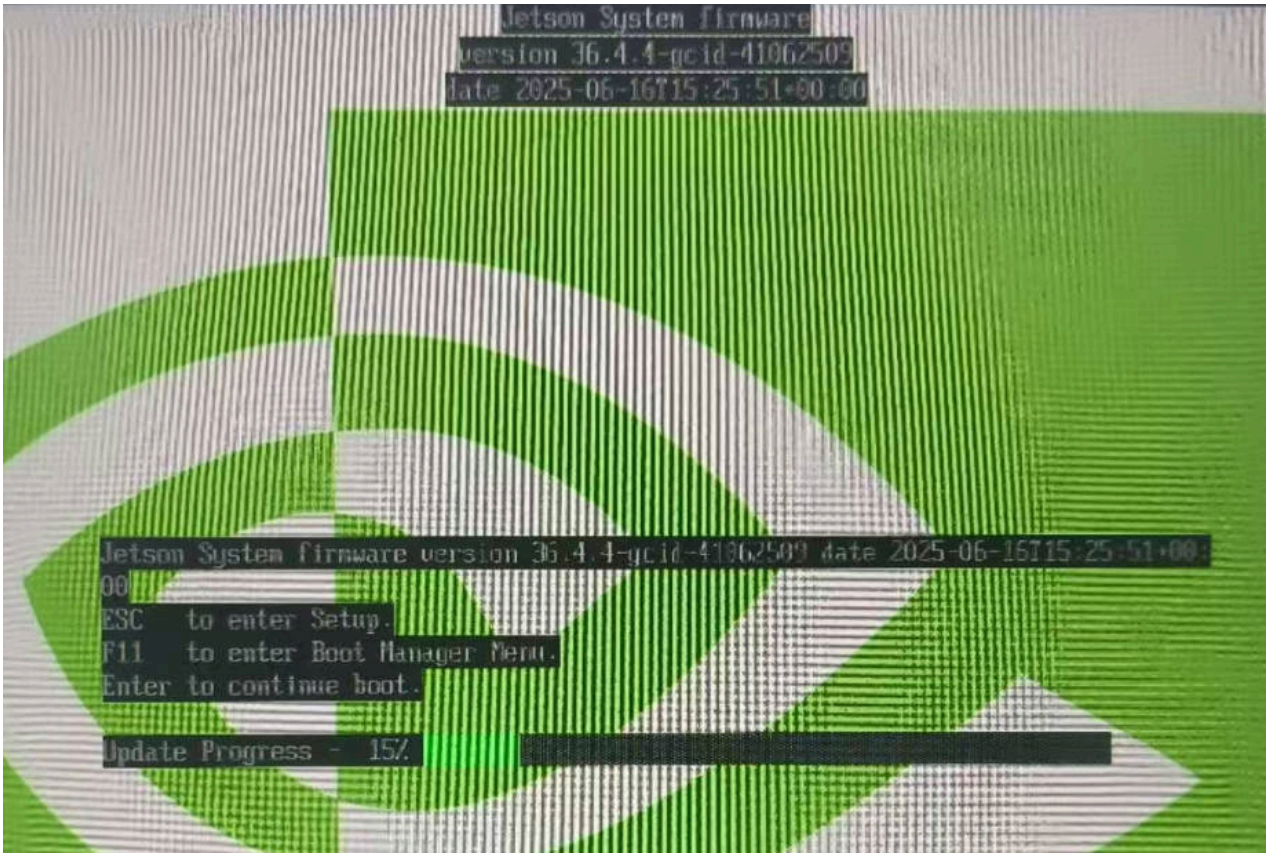




3. If `NVIDIA QSPI Firmware Update Required` appears, press `Y` **before the 30-second countdown expires**, and wait for both QSPI updates. The device may reboot on its own during this time — do not cut the power; if it does not return to the USB installer automatically after rebooting, select the USB drive again from `Boot Manager`. If the prompt does not appear, the installer is not requesting an

update this time — continue with the remaining installation steps, and do not reinstall repeatedly just to trigger the prompt.







4. If the QSPI update prompt appeared in step 3, wait until both updates are complete and the system has returned to the boot flow, then confirm that the firmware version on the boot screen is no longer the old `36.4.4` (if it was a different version to begin with, confirm that the version has been updated as required by the installer). If the old version is still shown, the update did not succeed — **stop here and do not continue with the NVMe installation** — and record any errors that occurred during the update. If the update prompt did not appear in step 3, there is no need to wait for a version change; go directly to the installation menu and choose `Install Jetson ISO r39.2.1` → `Install on NVMe`, then verify the target SSD before confirming. The installer automatically partitions, formats and writes the system — you do not need to format the SSD manually, and the old data will be erased. Wait until the installer explicitly reports completion and prompts you to reboot; do not judge success by an automatic reboot or a brief black screen.
5. After the installation finishes, remove the installation USB drive as prompted and boot from the NVMe.

If the system has already rebooted by itself while the USB drive is still inserted, shut down normally, remove the drive, and then boot again. On the first boot, follow the `oem-config` prompts to set up the language, network and account. If the system hangs on the underscore in the top-left corner, returns to the installer, or fails to enter the desktop, do not repeatedly re-flash the NVMe: first check the UEFI firmware version and the installer's final success/failure message.



4 Acceptance Check

Once the new system has booted, run:

```
cat /etc/nv_tegra_release
cat /etc/os-release
findmnt /
lsblk -o NAME,SIZE,MODEL,MOUNTPOINTS
sudo nvbootctrl dump-slots-info
```

Confirm that the QSPI/UEFI that needed updating has been updated, that the L4T version is R39.2.1, that the root filesystem is indeed on the expected NVMe, and that the device can boot on its own again after the USB drive is removed. The firmware version on the boot screen and the system version in `/etc/nv_tegra_release` come from the QSPI and the NVMe respectively, so they cannot substitute for each other during acceptance. This ISO installs the BSP/operating system first; whether the JetPack SDK components such as CUDA and TensorRT are complete must be verified separately, and they should be installed by referring to the NVIDIA JetPack download page. A successful system boot alone is not enough to conclude that the component environment has been restored.

Official references: JetPack 7.2.1 download notes, QSPI update and ISO installation steps for the Orin Nano Developer Kit, Installing Jetson Linux R39.2.1 from a USB ISO.

Installing the Official Clean System in a Virtual Machine

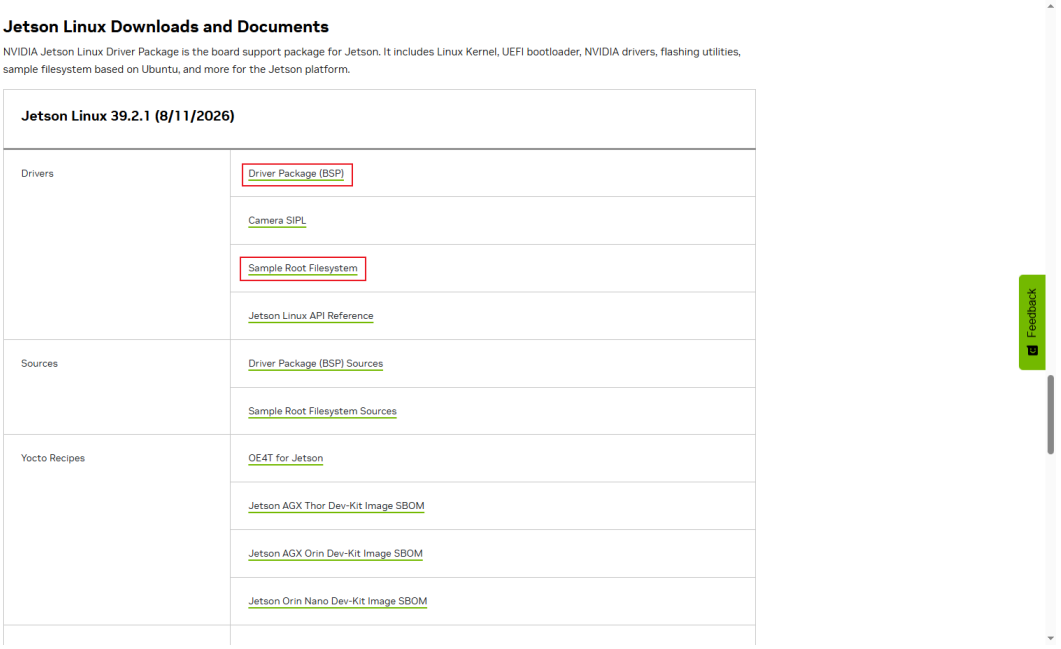
Requires an Ubuntu 22.04 host, Recovery mode and a Type-C data cable

1 Download the Files

Official website: https://developer.nvidia.com/embedded/jetpack/downloads/archive-7.2.1

Note: NVIDIA Jetson Linux 39.2.1 corresponds to JetPack 7.2.1

Download the compressed archives for `Driver Package (BSP)` and `Sample Root Filesystem`:



Jetson Linux Downloads and Documents

NVIDIA Jetson Linux Driver Package is the board support package for Jetson. It includes Linux Kernel, UEFI bootloader, NVIDIA drivers, flashing utilities, sample filesystem based on Ubuntu, and more for the Jetson platform.

Jetson Linux 39.2.1 (8/11/2026)	
Drivers	Driver Package (BSP)
	Camera SIPL
	Sample Root Filesystem
	Jetson Linux API Reference
Sources	Driver Package (BSP) Sources
	Sample Root Filesystem Sources
Yocto Recipes	OE4T for Jetson
	Jetson AGX Thor Dev-Kit Image SBOM
	Jetson AGX Orin Dev-Kit Image SBOM
	Jetson Orin Nano Dev-Kit Image SBOM

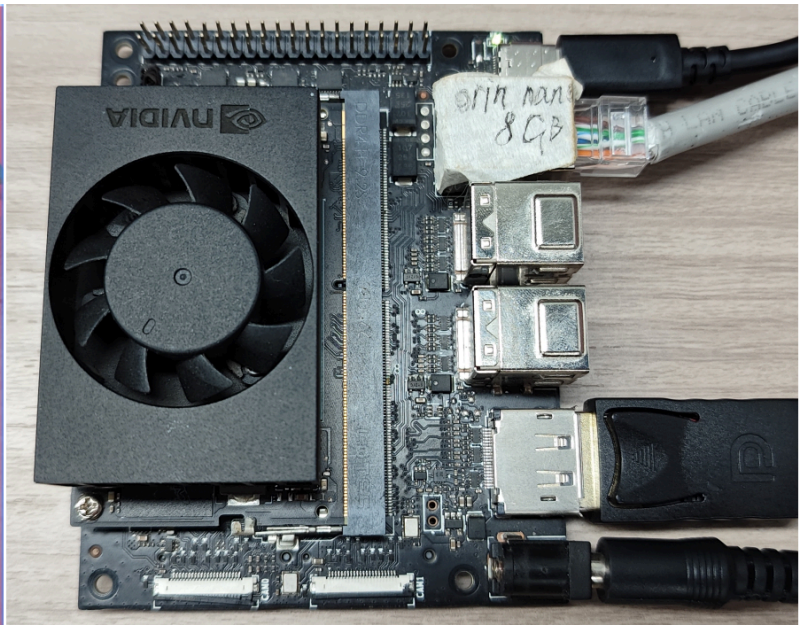
Feedback

2 Flash Mode

2.1 Hardware Connections

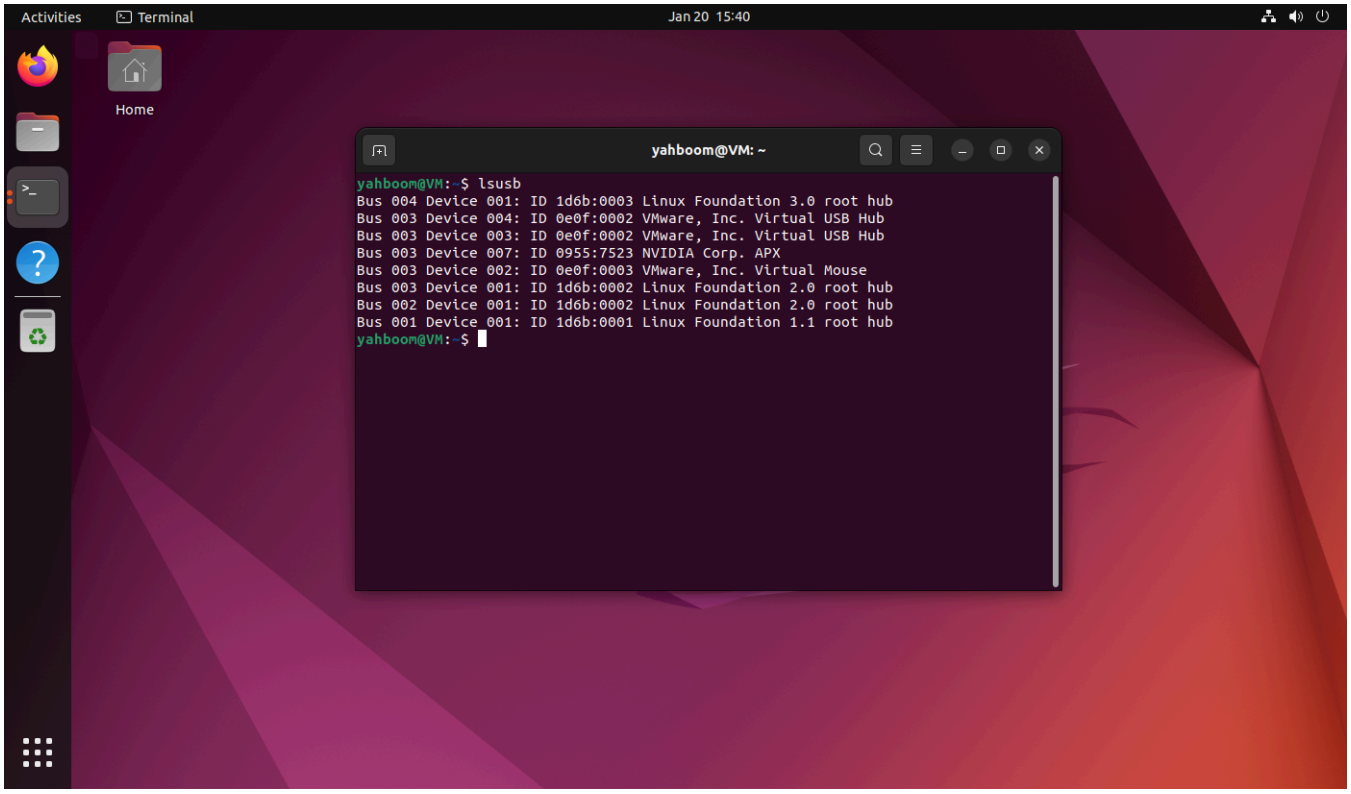
1. Use a jumper cap to short the FC REC and GND pins below the core board: you do not need to remove the core board — the picture is only there to make the observation clearer.
2. The Jetson Orin board needs a DC power adapter, a DP cable, an Ethernet cable and a Type-C data cable: connect the Type-C data cable to your computer.





2.2 Software Connection

Once the board has connected to the Ubuntu system successfully, the `lsusb` command shows the `NVIDIA Corp. APX` entry:



> Note: make sure the virtual machine has enough free space before you start

3 Flash the System

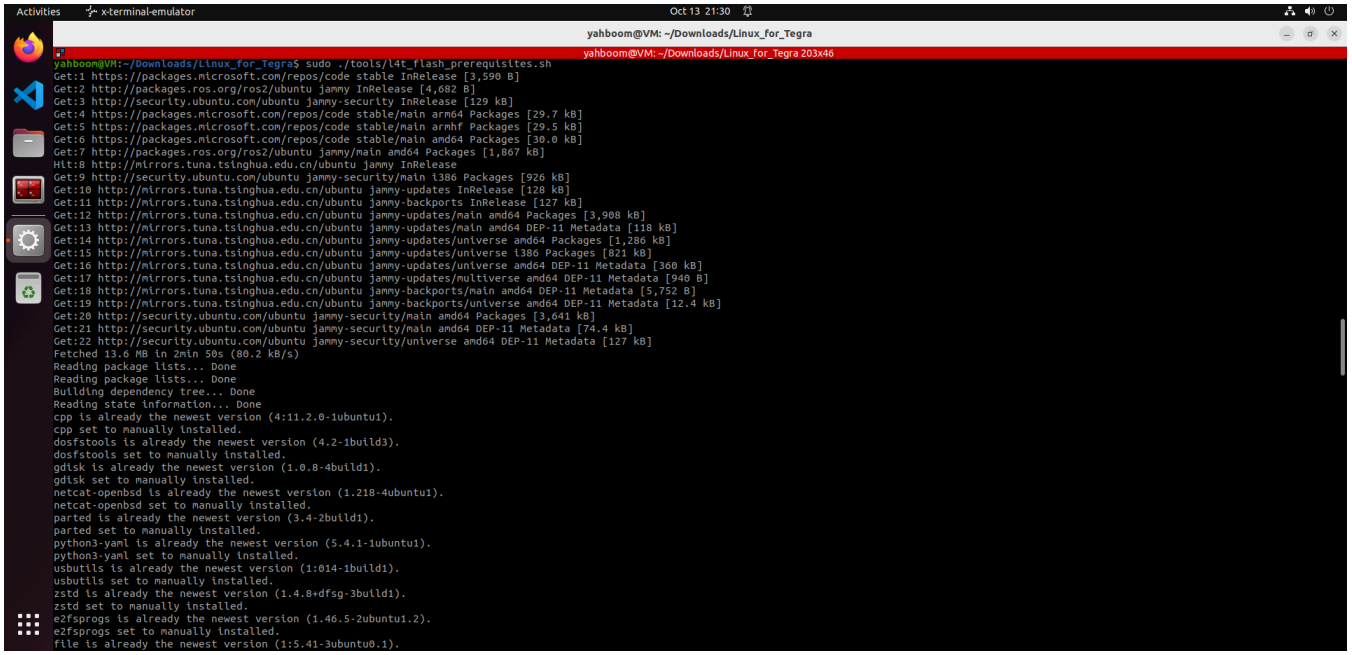
3.1 Extract the Files

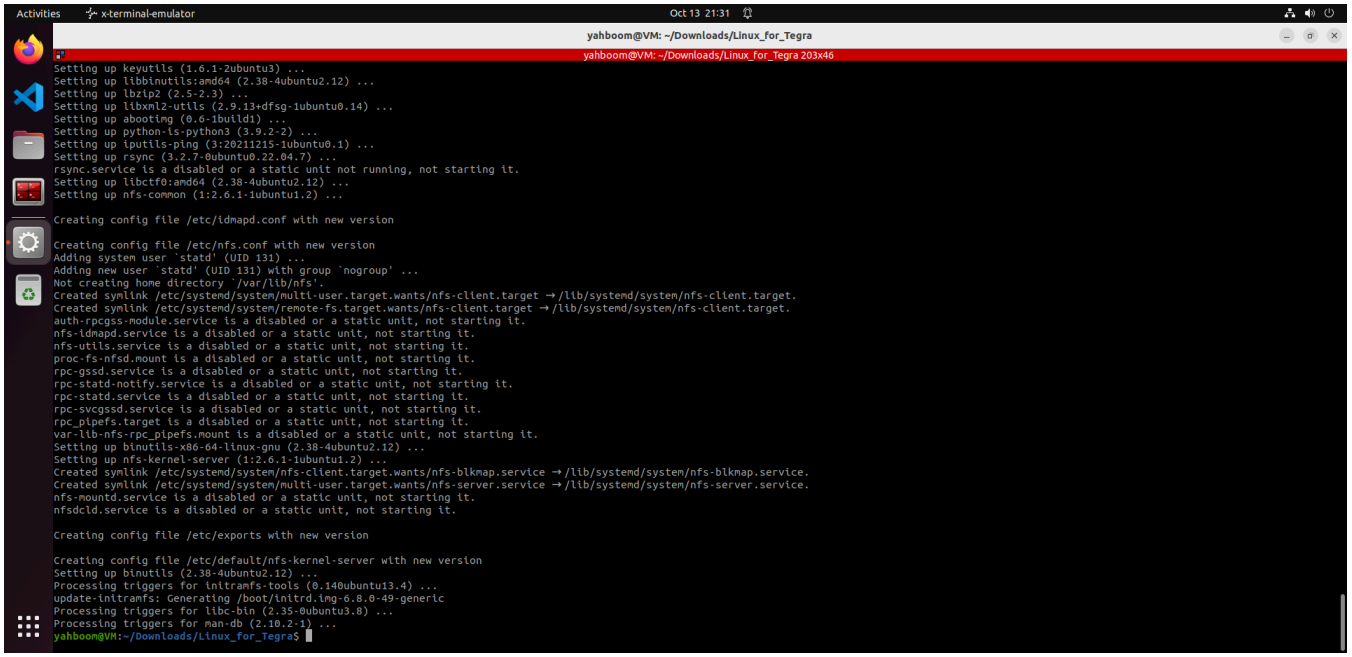
Open a terminal in the download folder, then extract the files in the terminal and change into the target folder:

```
tar xvf Jetson_Linux_R39.2.1_aarch64.tbz2
sudo tar -xpvf Tegra_Linux_Sample-Root-Filesystem_R39.2.1_aarch64.tbz2 -C
Linux_for_Tegra/rootfs/
cd Linux_for_Tegra/
```

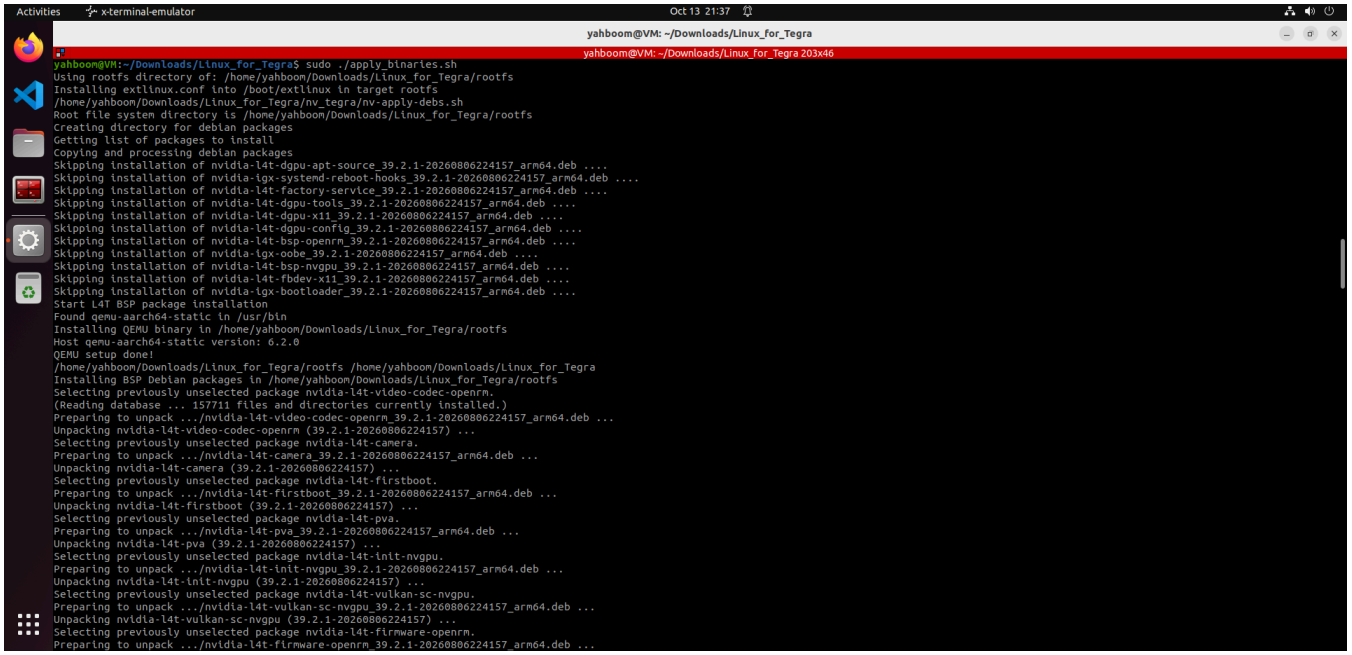
3.2 Run the Scripts

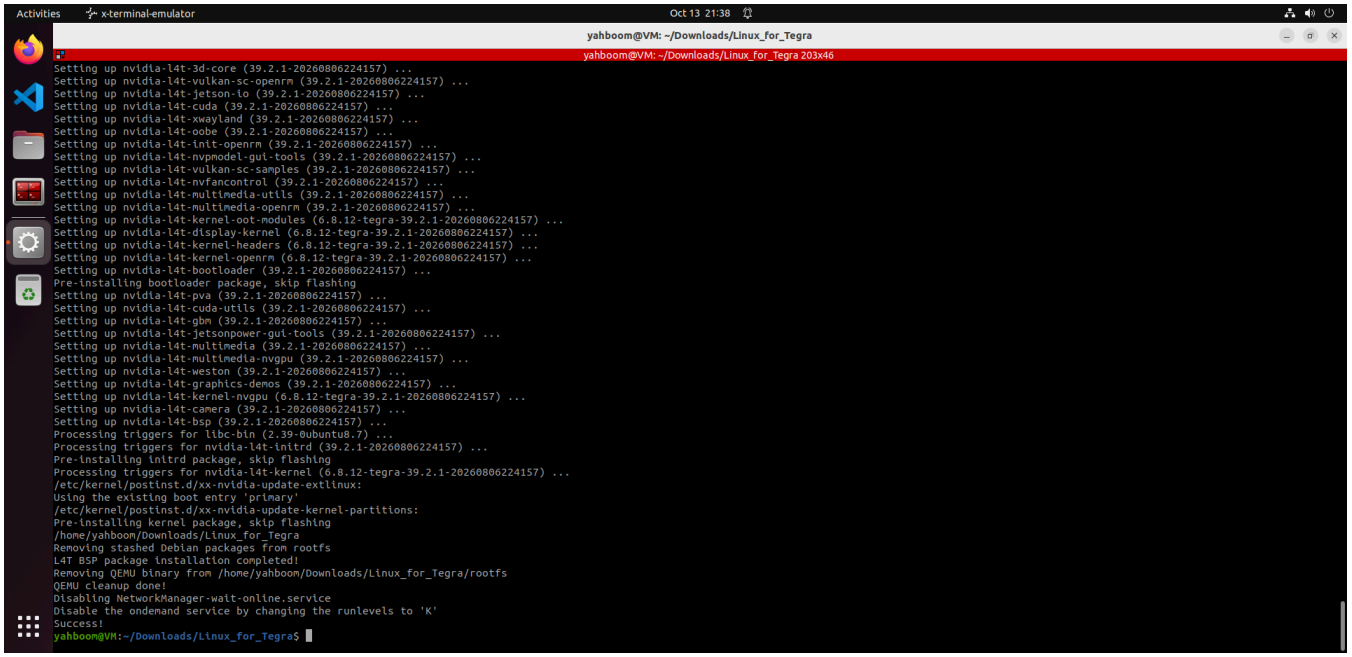
```
sudo ./tools/l4t_flash_prerequisites.sh
```





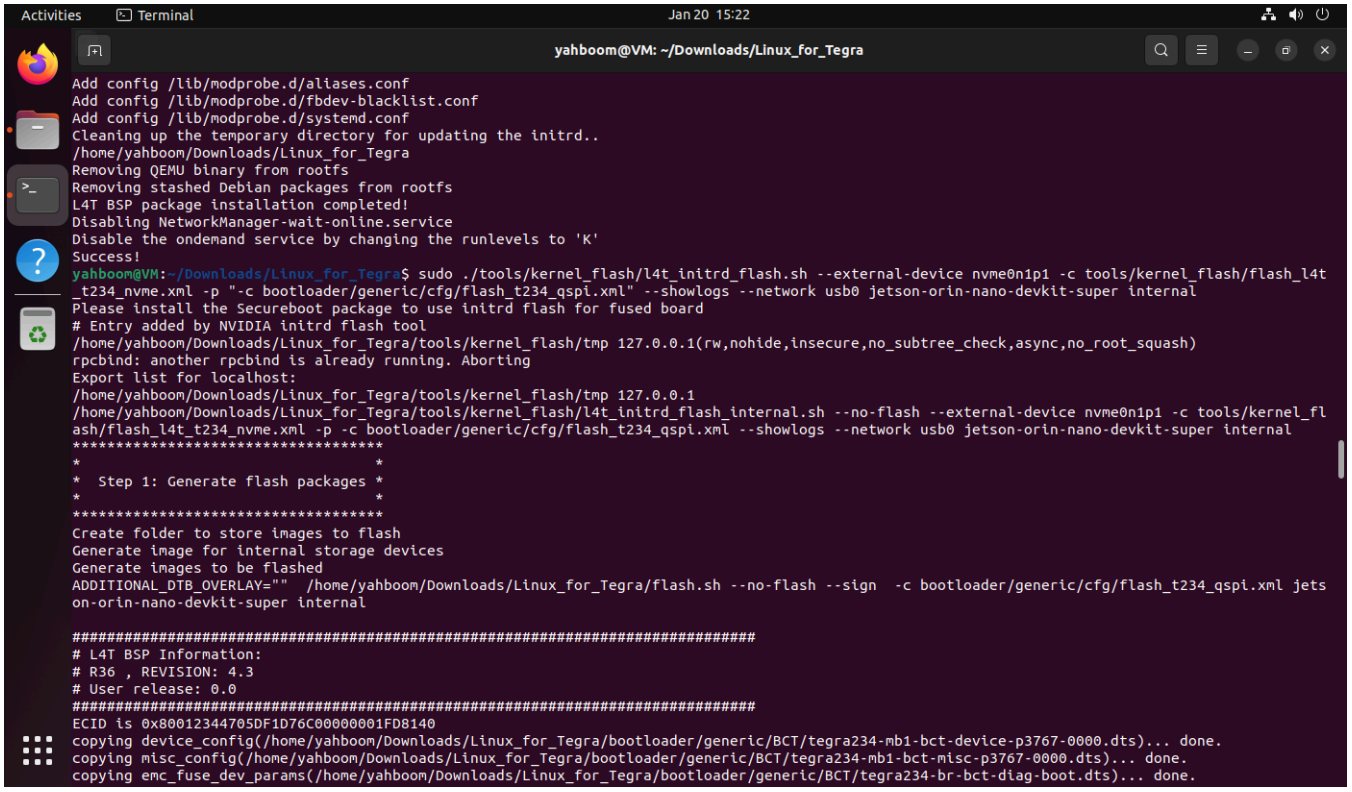

```
sudo ./apply_binaries.sh
```

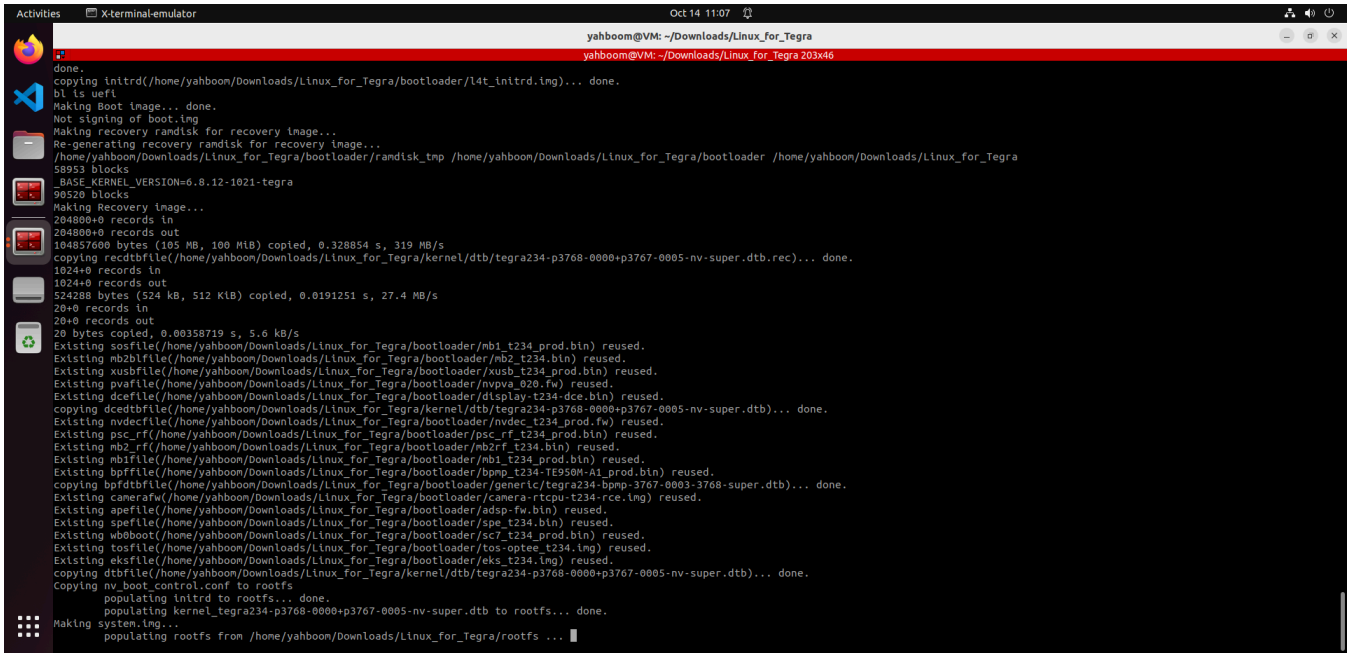




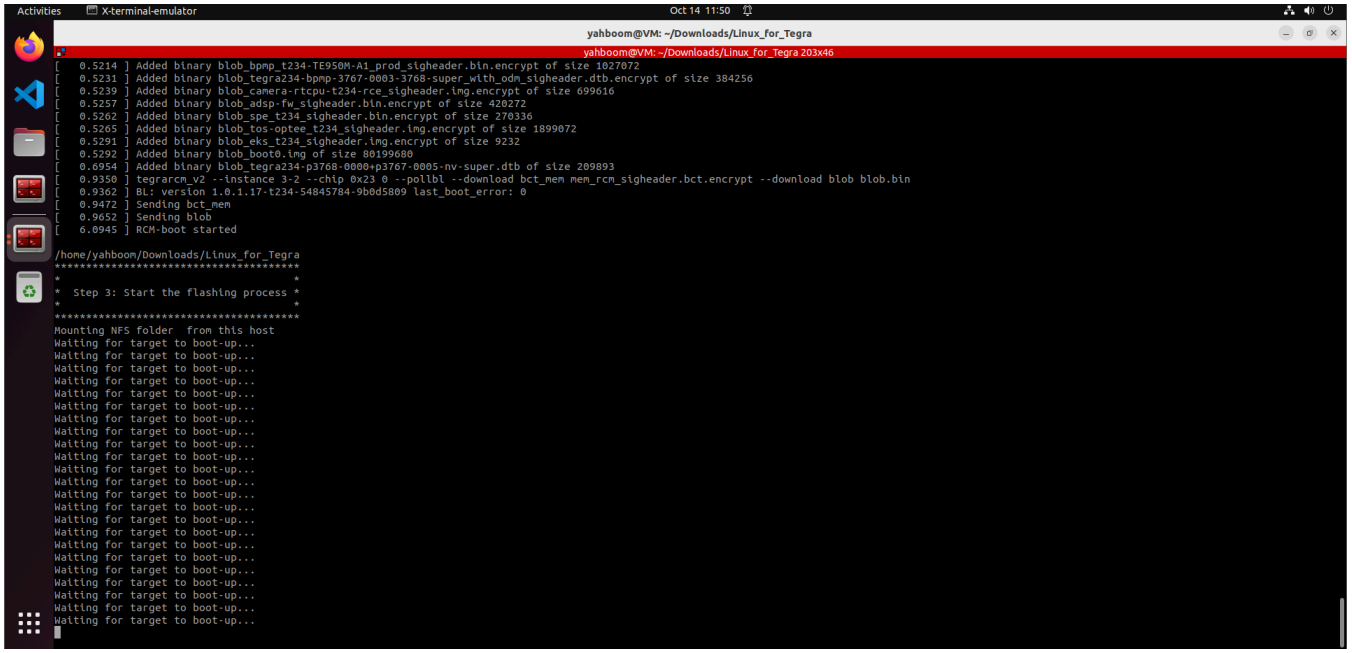
3.3 Flash the System to the SSD

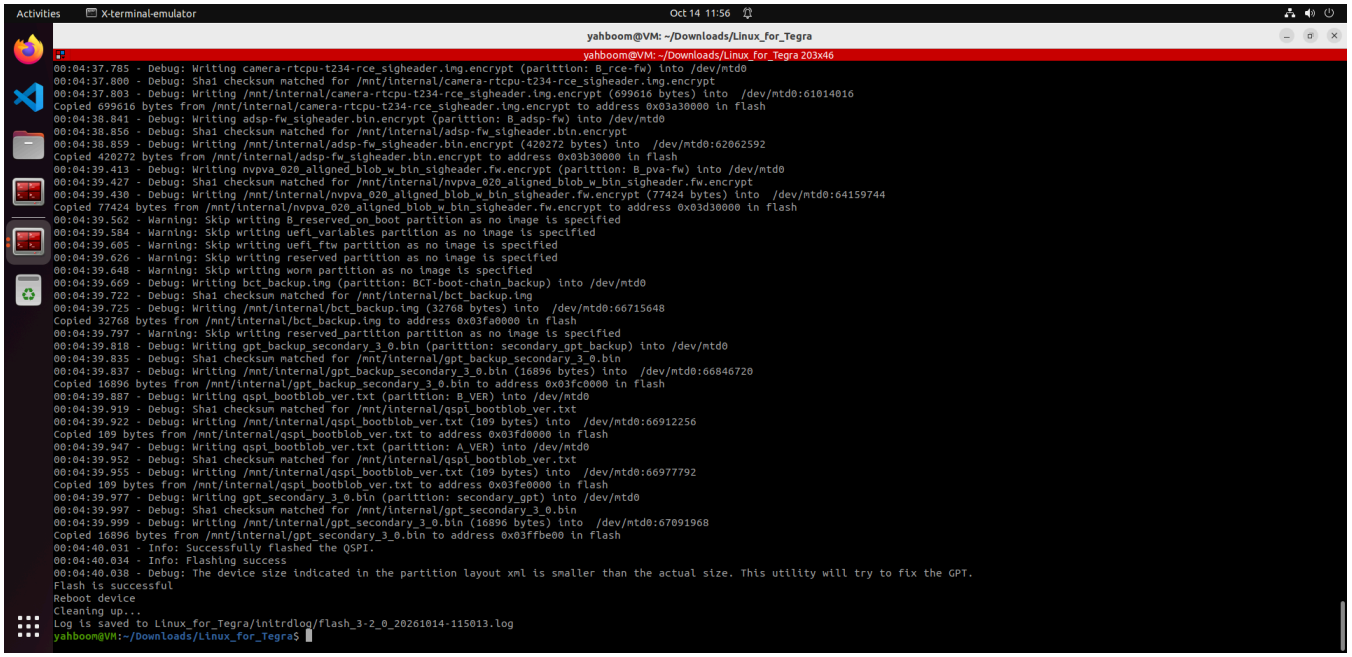
```
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device nvme0n1p1 -c
tools/kernel_flash/flash_l4t_t234_nvme.xml -p "-c
bootloader/generic/cfg/flash_t234_qspi.xml" --showlogs --network usb0 jetson-orin-nano-
devkit-super internal
```





> Note: During flashing you must reconnect the device to the virtual machine in time, otherwise the connection will time out!

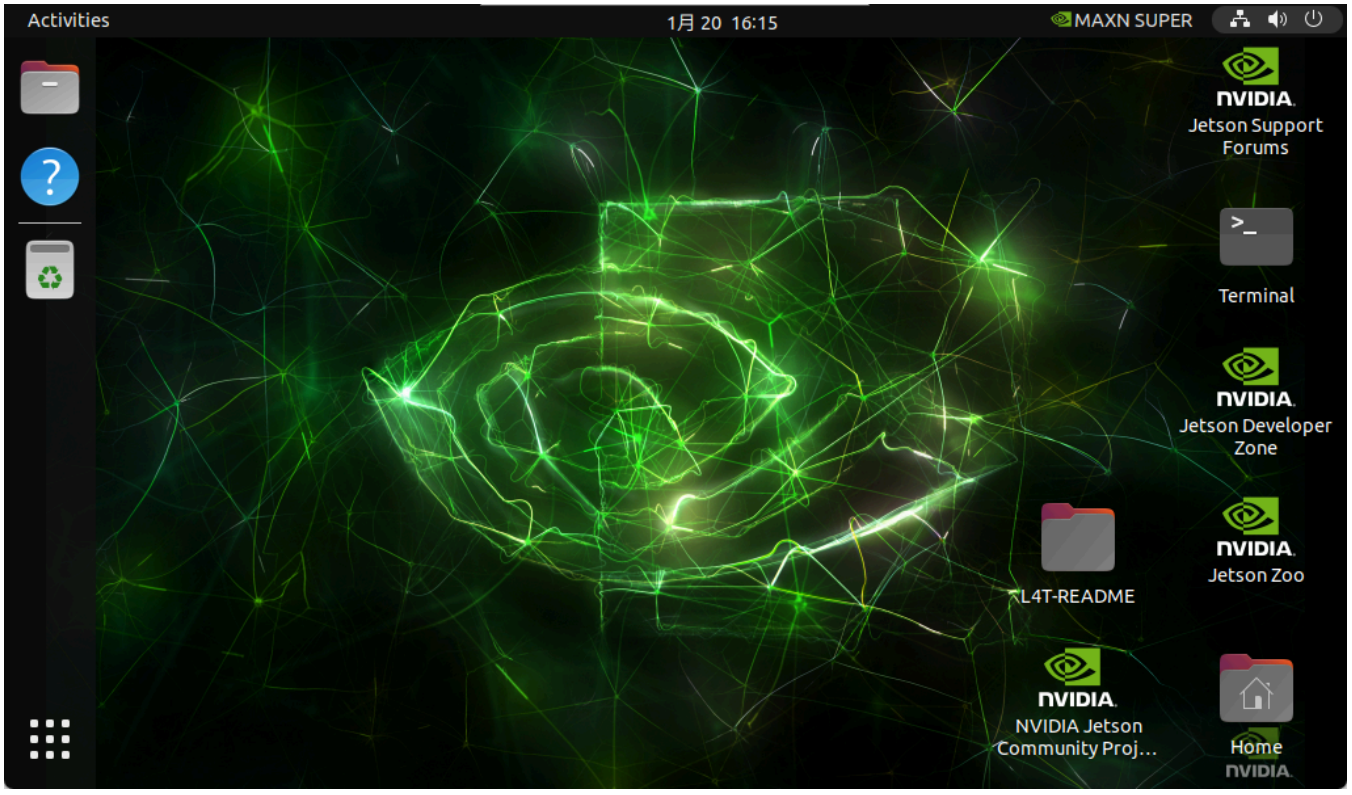




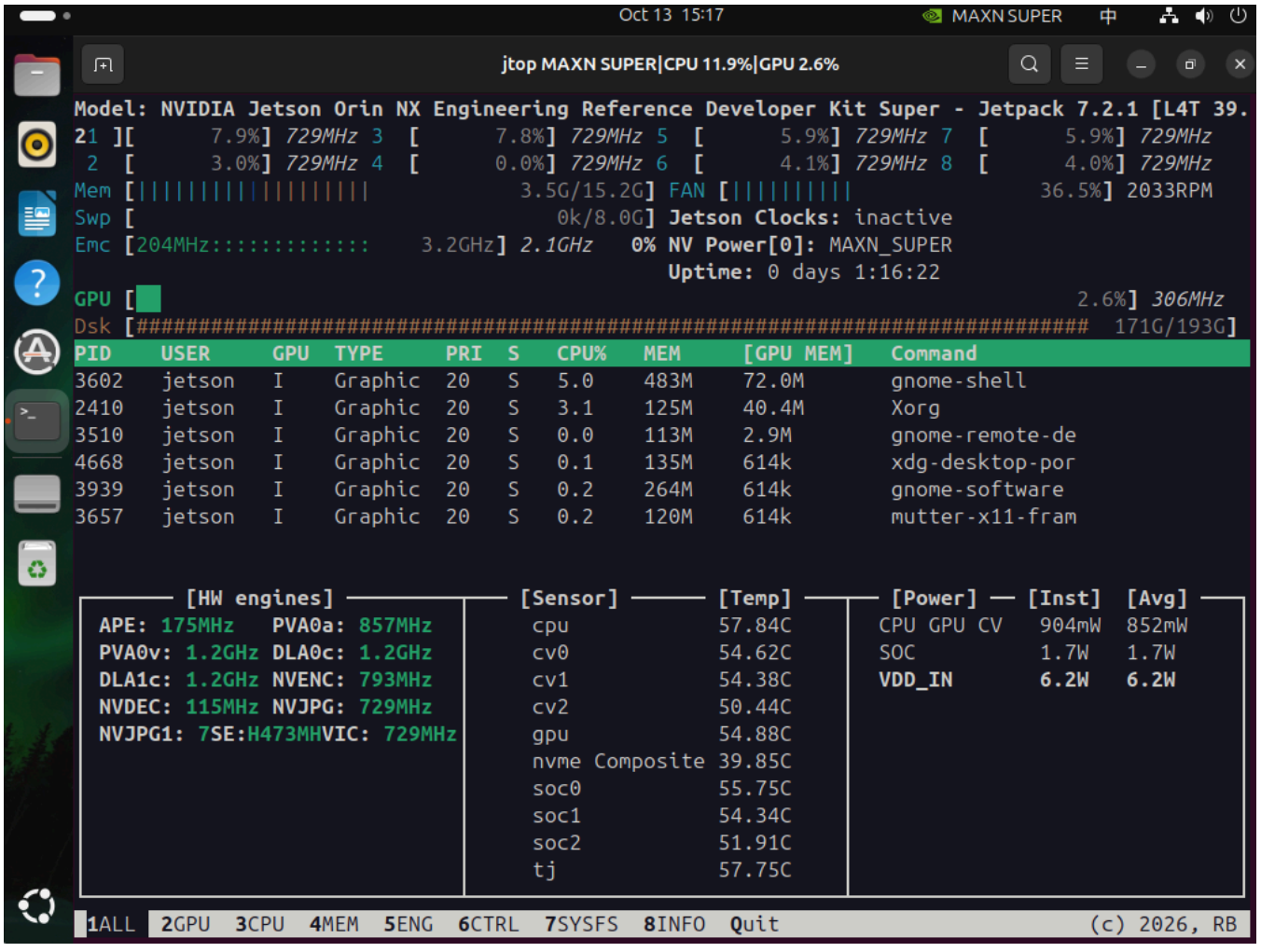
4 Boot the System

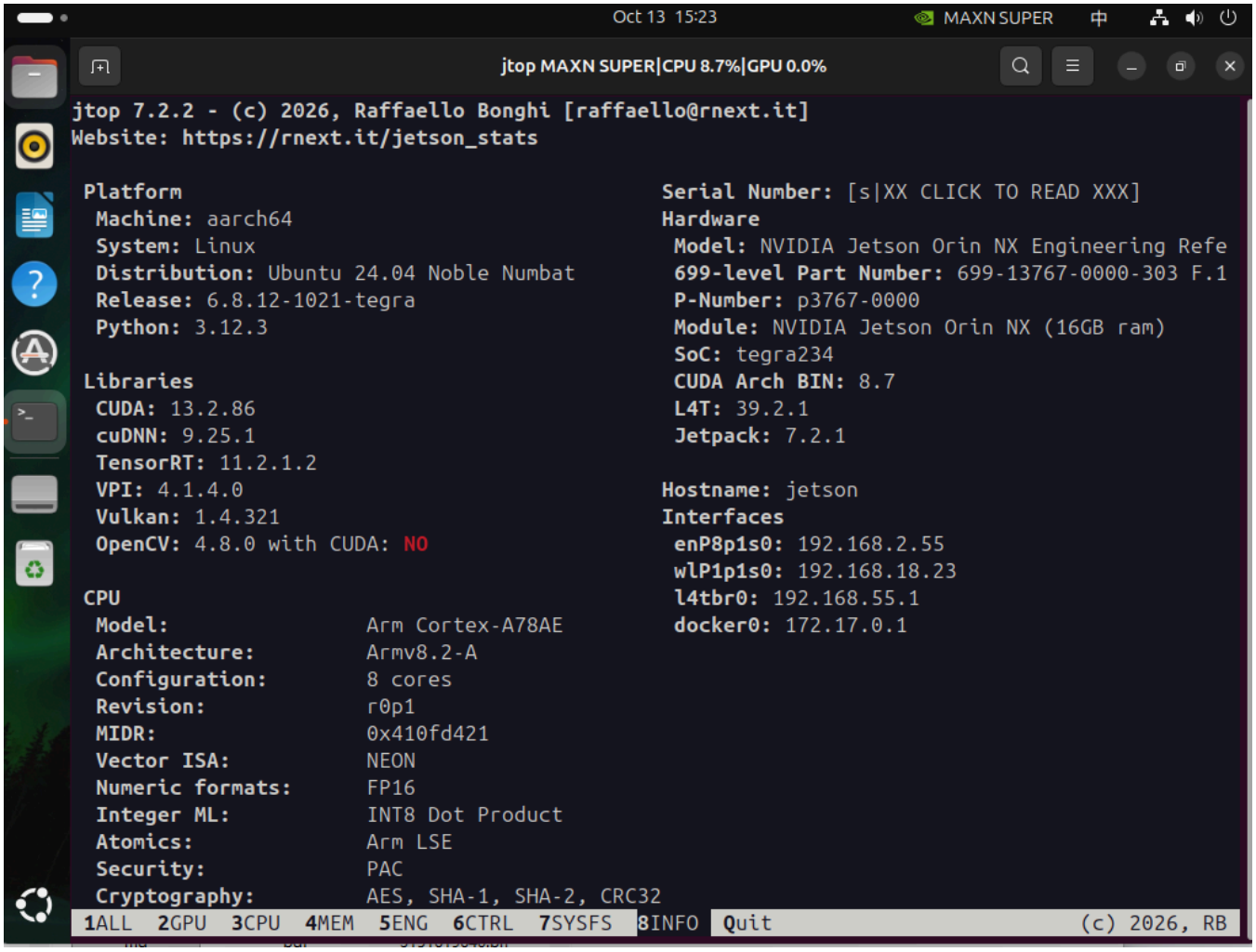
After the system has been flashed successfully, disconnect the board's power (unplug the DC power adapter and the Type-C data cable), then remove the jumper cap that shorts FC REC and GND below the core board. After confirming that these steps are done, reconnect the DC power adapter and the DP cable (connected to the display), and boot the system.

Enter the time zone, username, password and other details as prompted, and then enter the system



The clean system does not include the jtop tool; refer to 【Chapter 3 Basic Usage → 08. Using the Jtop Tool】 for how to install and use it





5 Component Environment

The above only covers flashing the clean system. If you need the `CUDA`, `TensorRT` and other environments, refer to 【Chapter 2 Board Basics: Installing the Jetson Component Environment】 as well!